

Follow today's class instruction (running Pinecone job as a Airflow job)

1. Modify your docker-compose.yaml file to include sentence-transformers and pinecone packages (1pt)

```
pip install sentence-transformers==3.1.1 pinecone==5.3.1
```

- Restart your docker containers ("docker compose down" and "docker compose up")

```
PS E:\SJSU\class\data226\airflow> docker compose up
[+] up 4/4
✓ Network airflow_default          Created      0.0s
✓ Container airflow-postgres-1     Created      0.1s
✓ Container airflow-airflow-init-1 Created      0.1s
✓ Container airflow-airflow-1      Created      0.1s
Attaching to airflow-1, airflow-init-1, postgres-1
Container airflow-postgres-1 Waiting
postgres-1 |
postgres-1 | PostgreSQL Database directory appears to contain a database; Skipping initialization
```

- docker-compose.yaml should be a part of your github submission
- If needed, pinned versions: sentence-transformers==3.1.1 pinecone==5.3.1

2. Configure Pinecone (account), get the API token and create Airflow Variable (1pt)

Pinecone / Matthew's Org / Default / API keys

Get started
Database
Assistant
Inference
API keys
Manage

API keys

Name	Created on	Created by	Value
default	4/30/2026	Matthew Chen	pcsk_4p4oN8_*****

Choose File No file chosen ☒ Overwrite if exists ☐ Fail if exists ☐ Skip if exists [Import Variables](#)

List Variable

Search

+ Actions

	Key	Val	Description
☐	pinecone_api_key	*****	

3. Download, Process and Generate an input file to Pinecone (2pt)

DAG: Medium_to_Pinecone Build a Medium Posting Search Engine using Pinecone

04/30/2026 09:35:25 AM All Run Types All Run States [Clear Filters](#)

Press **SHIFT** + **F** for Shortcuts

Duration: 00:01:59

download_data
preprocess_data
create_pinecone_index
generate_embeddings_and_upsert
test_search_query

DAG: Medium_to_Pinecone Run: 2026-04-23, 09:33:05 UTC Task: download_data

Details Graph Gantt Code Event Log **Logs** XCom Task Duration

All Levels All File Sources

```
c9019daff6f1
*** Found local files:
*** /opt/airflow/logs/dag_id=Medium_to_Pinecone/run_id=manual_2026-04-30T09:33:05.894673+08:00/task_id=download_data/attempt=1.log
[2026-04-30, 09:33:09 UTC] [local_task_job_runner.py:123] ▶ Pre task execution logs
[2026-04-30, 09:33:10 UTC] [logging_mixin.py:190] INFO - Downloaded file has 2459 lines
[2026-04-30, 09:33:10 UTC] [python.py:240] INFO - Done. Returned value was: /tmp/medium_data/medium_data.csv
[2026-04-30, 09:33:10 UTC] [taskinstance.py:340] ▶ Post task execution logs
```

Airflow DAGs Cluster Activity Datasets Security Browse Admin Docs

DAG: Medium_to_Pinecone Build a Medium Posting Search Engine using Pinecone

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Press **shift** + **/** for Shortcuts

Duration

00 01 59

00 00 59

00 00 00

download_data

preprocess_data

create_pinecone_index

generate_embeddings_and_upsert

test_search_query

DAG: Medium_to_Pinecone Run Task: preprocess_data

2026-04-23, 09:33:05 UTC

Details Graph Gantt Code Event Log **Logs** XCom Task Duration

All Levels All File S

```

cf019daff6f1
*** Found local files:
*** /opt/airflow/logs/dag_id=Medium_to_Pinecone/run_id=manual_2026-04-30T09:33:05.894673+00:00/task_id=preprocess_data/attempt=1.log
[2026-04-30, 09:33:14 UTC] (local_task_job_runner.py:123) ▶ Pre task execution logs
[2026-04-30, 09:33:14 UTC] (logging_mixin.py:190) INFO - Preprocessed data saved to /tmp/medium_data/medium_preprocessed.csv
[2026-04-30, 09:33:14 UTC] (python.py:240) INFO - Done. Returned value was: /tmp/medium_data/medium_preprocessed.csv
[2026-04-30, 09:33:14 UTC] (taskinstance.py:340) ▶ Post task execution logs

```

4. Create Pinecone index (1tp)

DAG: Medium_to_Pinecone Build a Medium Posting Search Engine using Pinecone

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Press **shift** + **/** for Shortcuts

Duration

00 01 59

00 00 59

00 00 00

download_data

preprocess_data

create_pinecone_index

generate_embeddings_and_upsert

test_search_query

DAG: Medium_to_Pinecone Run Task: create_pinecone_index

2026-04-23, 09:33:05 UTC

Details Graph Gantt Code Event Log **Logs** XCom Task Duration

All Levels All File Sources

```

cf019daff6f1
*** Found local files:
*** /opt/airflow/logs/dag_id=Medium_to_Pinecone/run_id=manual_2026-04-30T09:33:05.894673+00:00/task_id=create_pinecone_index/attempt=1.log
[2026-04-30, 09:33:09 UTC] (local_task_job_runner.py:123) ▶ Pre task execution logs
[2026-04-30, 09:33:10 UTC] (discover_namespace_packages.py:12) INFO - Discovering subpackages in NamespacePath(['/home/***/.local/lib/python3.12/site-packages/pinecone_plugins'])
[2026-04-30, 09:33:10 UTC] (discover_plugins.py:9) INFO - Looking for plugins in pinecone_plugins.inference
[2026-04-30, 09:33:10 UTC] (installation.py:10) INFO - Installing plugin inference into Pinecone
[2026-04-30, 09:33:20 UTC] (logging_mixin.py:190) INFO - Pinecone index 'semantic-search-fast' created successfully
[2026-04-30, 09:33:20 UTC] (python.py:240) INFO - Done. Returned value was: semantic-search-fast
[2026-04-30, 09:33:20 UTC] (taskinstance.py:340) ▶ Post task execution logs

```

5. Convert the input file into embeddings and ingest them into Pinecone (2pt)

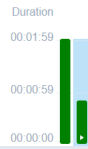
04/30/2026

09:35:25 AM

All Run Types All Run States

[Clear Filters](#)

Press **shift** + **/** for Shortcuts



download_data	■ ■
preprocess_data	■ ■
create_pinecone_index	■ ■
generate_embeddings_and_upsert	■ ■
test_search_query	■ ■

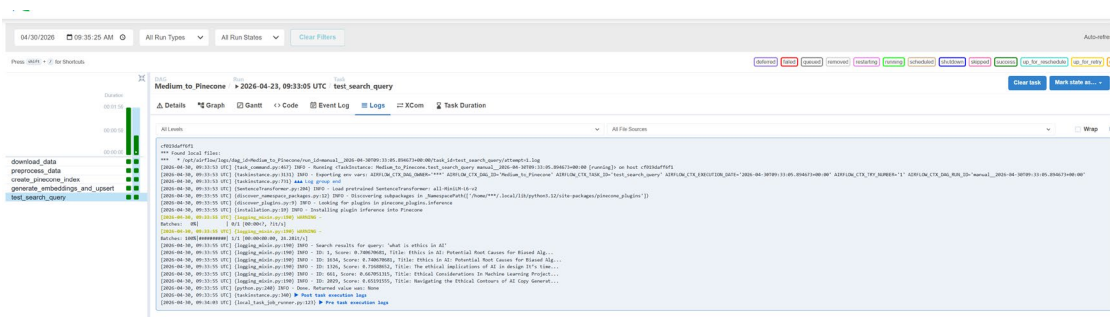
DAG **Run** **Task**
Medium_to_Pinecone / ▶ **2026-04-23, 09:33:05 UTC** / **generate_embeddings_and_upsert**

[Details](#)
[Graph](#)
[Gantt](#)
[Code](#)
[Event Log](#)
[Logs](#)
[XCom](#)
[Task Duration](#)

All Levels

```
[2026-04-30, 09:33:45 UTC] [logging_mixin.py:190] WARNING -
Batches: 0% | 0/4 [00:00:?, ?it/s]
[2026-04-30, 09:33:45 UTC] [logging_mixin.py:190] INFO - Processing batch 20/25
[2026-04-30, 09:33:45 UTC] [logging_mixin.py:190] WARNING -
Batches: 0% | 0/4 [00:00:?, ?it/s]
[2026-04-30, 09:33:45 UTC] [logging_mixin.py:190] WARNING -
Batches: 25%|#5 | 1/4 [00:00:00:00, 9.12it/s]
[2026-04-30, 09:33:45 UTC] [logging_mixin.py:190] WARNING -
Batches: 75%|##### | 3/4 [00:00:00:00, 14.42it/s]
[2026-04-30, 09:33:45 UTC] [logging_mixin.py:190] WARNING -
Batches: 100%|##### | 4/4 [00:00:00:00, 17.29it/s]
[2026-04-30, 09:33:46 UTC] [logging_mixin.py:190] INFO - Processing batch 21/25
[2026-04-30, 09:33:46 UTC] [logging_mixin.py:190] WARNING -
Batches: 0% | 0/4 [00:00:?, ?it/s]
[2026-04-30, 09:33:46 UTC] [logging_mixin.py:190] WARNING -
Batches: 25%|#5 | 1/4 [00:00:00:00, 4.02it/s]
[2026-04-30, 09:33:46 UTC] [logging_mixin.py:190] WARNING -
Batches: 75%|##### | 3/4 [00:00:00:00, 9.29it/s]
[2026-04-30, 09:33:46 UTC] [logging_mixin.py:190] WARNING -
Batches: 100%|##### | 4/4 [00:00:00:00, 10.65it/s]
[2026-04-30, 09:33:47 UTC] [logging_mixin.py:190] INFO - Processing batch 22/25
[2026-04-30, 09:33:47 UTC] [logging_mixin.py:190] WARNING -
Batches: 0% | 0/4 [00:00:?, ?it/s]
[2026-04-30, 09:33:47 UTC] [logging_mixin.py:190] WARNING -
Batches: 25%|#5 | 1/4 [00:00:00:00, 4.39it/s]
[2026-04-30, 09:33:47 UTC] [logging_mixin.py:190] WARNING -
Batches: 75%|##### | 3/4 [00:00:00:00, 9.46it/s]
[2026-04-30, 09:33:47 UTC] [logging_mixin.py:190] WARNING -
Batches: 100%|##### | 4/4 [00:00:00:00, 10.95it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] INFO - Processing batch 23/25
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 0% | 0/4 [00:00:?, ?it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 25%|#5 | 1/4 [00:00:00:00, 6.91it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 75%|##### | 3/4 [00:00:00:00, 12.03it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 100%|##### | 4/4 [00:00:00:00, 14.45it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] INFO - Processing batch 24/25
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 0% | 0/4 [00:00:?, ?it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] INFO - Processing batch 25/25
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 0% | 0/4 [00:00:?, ?it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 25%|#5 | 1/4 [00:00:00:00, 6.06it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 75%|##### | 3/4 [00:00:00:00, 12.10it/s]
[2026-04-30, 09:33:48 UTC] [logging_mixin.py:190] WARNING -
Batches: 100%|##### | 4/4 [00:00:00:00, 14.10it/s]
[2026-04-30, 09:33:49 UTC] [logging_mixin.py:190] WARNING -
Batches: 25%|#5 | 1/4 [00:00:00:00, 5.81it/s]
[2026-04-30, 09:33:49 UTC] [logging_mixin.py:190] WARNING -
Batches: 75%|##### | 3/4 [00:00:00:00, 9.86it/s]
[2026-04-30, 09:33:49 UTC] [logging_mixin.py:190] WARNING -
Batches: 100%|##### | 4/4 [00:00:00:00, 12.04it/s]
[2026-04-30, 09:33:49 UTC] [logging_mixin.py:190] INFO - Successfully upserted 2498 records to Pinecone
[2026-04-30, 09:33:49 UTC] [python:240] INFO - Done. Returned value was: semantic-search-fast
[2026-04-30, 09:33:49 UTC] [taskinstance.py:340] ▶ Post task execution logs
```

6. Run search against Pinecone (1pt)



semantic-search-fast

Record count 2,498 Region us-east-1 Type Dense Host semantic-search-fast-4dkeyf-svc.aped-4d27-374a...

[BROWSER](#) [METRICS](#) [NAMESPACES \(1\)](#) [CONFIGURATION](#)

Records

Namespace	Operation	ID
__default__	Search by ID	0

[+ Filter](#) [+ Rerank](#)

Search: 10 results (top_k=10)

1

_id: 1633
title: "Not All Rainbows and Sunshine: The Darker Side of ChatGPT Part 1: The Risks and Ethical Issues..."

SCORE
1.0009

2

_id: 0
title: "Not All Rainbows and Sunshine: The Darker Side of ChatGPT Part 1: The Risks and Ethical Issues..."

SCORE
1.0009

3

_id: 1665
title: "ChatGPT—Handle With Care Behind the Hype—Understanding what ChatGPT can do and what it cannot do..."

SCORE
0.5685

4

_id: 975
title: "ChatGPT—Handle With Care Behind the Hype—Understanding what ChatGPT can do and what it cannot do..."

SCORE
0.5685

DAG: Medium_to_Pinecone Build a Medium Posting Search Engine using Pinecone

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Press **shift** + **/** for Shortcuts

Duration

00:01:59

00:00:59

00:00:00

download_data

preprocess_data

create_pinecone_index

generate_embeddings_and_upsert

test_search_query

DAG: Medium_to_Pinecone / Run

2026-04-23, 09:33:05 UTC

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download_data
success
@task

preprocess_data
success
@task

create_pinecone_index
success
@task

generate_embeddings_and_upsert
success
@task

test_search_query
success
@task

- For each step, capture screenshots accordingly
 1. For Airflow tasks, capture the logs screen of each task
- Create a github repo for this and add the link to the report
https://github.com/MaisuC/pinecone_airflow